

Mathematical Statistics

Final Cheat Sheet

Weeks 9-14: Functions of RVs, Sampling,
Estimation & Testing
Xuang Liang
xl3493@cumc.columbia.edu

1. Functions of Random Variables

Method 1: CDF Method (Univariate)

For $Y = g(X)$:

- Find CDF:
 $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$
- Differentiate: $f_Y(y) = F'_Y(y)$

Method 2: Transformation Formula

If $Y = g(X)$ is monotonic and differentiable:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right| = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

For many-to-one (e.g., $Y = X^2$), sum over roots:

$$f_Y(y) = \sum_i f_X(x_i(y)) \left| \frac{dx_i}{dy} \right|$$

Method 3: MGF Method

If $M_Y(t) = M_Z(t)$, then $Y \stackrel{d}{=} Z$ (same distribution).

Method 4: Jacobian (Bivariate)

For $(U, V) = g(X, Y)$ with inverse
 $(X, Y) = h(U, V)$:

$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v)) |J|$$

Jacobian: $J = \det \left[\frac{\partial(x,y)}{\partial(u,v)} \right]$

Convolution (Sum of Independent RVs)

If X, Y independent, $Z = X + Y$:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

Discrete: $p_Z(z) = \sum_x p_X(x) p_Y(z-x)$

Order Statistics

For i.i.d. $X_1, \dots, X_n$ with CDF F and PDF f :

Minimum: $X_{(1)} = \min\{X_1, \dots, X_n\}$

$$F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n$$

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1} f(x)$$

Maximum: $X_{(n)} = \max\{X_1, \dots, X_n\}$

$$F_{X_{(n)}}(x) = [F(x)]^n$$

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1} f(x)$$

k-th order statistic:

$$f_{X_{(k)}}(x) \propto [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)$$

Uniform(0, 1): $X_{(k)} \sim \text{Beta}(k, n-k+1)$

Key Transformation Results

If $Z \sim N(0, 1)$:

- $Z^2 \sim \chi^2(1)$
- $\sum_{i=1}^n Z_i^2 \sim \chi^2(n)$ (independent Z_i)

If $X \sim \text{Uniform}(0, 1)$: $Y = -\log(X) \sim \text{Exp}(1)$

Box-Muller: If $U_1, U_2 \sim U(0, 1)$ indep, then

$X = \sqrt{-2 \ln U_1} \cos(2\pi U_2)$,

$Y = \sqrt{-2 \ln U_1} \sin(2\pi U_2)$ are independent $N(0, 1)$

2. Sampling Distributions

Law of Large Numbers (LLN)

If $X_1, X_2, \dots$ i.i.d. with $\mathbb{E}[X_i] = \mu$,
 $\text{Var}(X_i) = \sigma^2 < \infty$:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu$$

Chebyshev proof: $P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2/n}{\varepsilon^2} \rightarrow 0$

Central Limit Theorem (CLT)

If X_i i.i.d. with mean μ and variance $\sigma^2 > 0$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

Equivalently: $\bar{X}_n \approx N(\mu, \sigma^2/n)$ for large n

For sums: $S_n = \sum X_i \approx N(n\mu, n\sigma^2)$

Continuity Correction (Discrete $\rightarrow$ Continuous)

For $X \sim B(n, p)$: $P(X \leq k) \approx P\left(Z \leq \frac{k+0.5-np}{\sqrt{np(1-p)}}\right)$

Rule: Normal approx reliable if $np(1-p) \gtrsim 10$

Chi-Square Distribution: $\chi^2(k)$

If $Z_1, \dots, Z_k \sim N(0, 1)$ independent:

$$\sum_{i=1}^k Z_i^2 \sim \chi^2(k)$$

PDF: $f(x) = \frac{1}{2^{k/2} \Gamma(k/2)} x^{k/2-1} e^{-x/2}, x > 0$

$\mathbb{E}[\chi^2(k)] = k, \text{Var}(\chi^2(k)) = 2k$

MGF: $M(t) = (1-2t)^{-k/2}, t < 1/2$

Additivity: If $X \sim \chi^2(k_1), Y \sim \chi^2(k_2)$ independent:

$$X + Y \sim \chi^2(k_1 + k_2)$$

t-Distribution: $t(n)$

If $Z \sim N(0, 1), V \sim \chi^2(n)$ independent:

$$T = \frac{Z}{\sqrt{V/n}} \sim t(n)$$

$\mathbb{E}[T] = 0 (n > 1), \text{Var}(T) = \frac{n}{n-2} (n > 2)$

As $n \rightarrow \infty: t(n) \rightarrow N(0, 1)$

F-Distribution: $F(n_1, n_2)$

If $V_1 \sim \chi^2(n_1), V_2 \sim \chi^2(n_2)$ independent:

$$F = \frac{V_1/n_1}{V_2/n_2} \sim F(n_1, n_2)$$

$\mathbb{E}[F] = \frac{n_2}{n_2-2} (n_2 > 2)$

If $F \sim F(n_1, n_2): \frac{1}{F} \sim F(n_2, n_1)$

If $T \sim t(n): T^2 \sim F(1, n)$

Normal Sample: Key Results

If $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ i.i.d.:

- $\bar{X} \sim N(\mu, \sigma^2/n)$
- $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$ where
 $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$
- $\bar{X}$ and S^2 are independent
- $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ (σ known)
- $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$ (σ unknown)

Delta Method

If $\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N(0, \tau^2)$ and $g'(\theta) \neq 0$:

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \Rightarrow N(0, [g'(\theta)]^2 \tau^2)$$

Variance: $\text{Var}(g(\hat{\theta}_n)) \approx [g'(\theta)]^2 \text{Var}(\hat{\theta}_n)$

3. Point Estimation

Bias and MSE

Bias: $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$

Unbiased: $\text{Bias}(\hat{\theta}) = 0$

Variance: $\text{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]$

MSE: $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$

Consistency

$\hat{\theta}_n$ is consistent if $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \rightarrow \infty$

Sufficient condition: If unbiased and

$\lim_{n \rightarrow \infty} \text{Var}(\hat{\theta}_n) = 0$

Or: If $\lim_{n \rightarrow \infty} \text{MSE}(\hat{\theta}_n) = 0$

Relative Efficiency

For two unbiased estimators $\hat{\theta}_1, \hat{\theta}_2$:

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

If $\text{eff} > 1$: $\hat{\theta}_1$ is more efficient (smaller variance)

Example: For $N(\mu, \sigma^2)$: $\text{eff}(\bar{X}, \bar{X}) = \frac{\pi}{2} \approx 1.57$
(mean vs median)

Method of Moments (MoM)

Equate sample moments to population moments:

- $\mu'_j = \mathbb{E}[X^j]$ (population moments)
- $m_j = \frac{1}{n} \sum X_i^j$ (sample moments)

3. Solve: $m_j = \mu'_j(\theta)$ for θ

Properties: Simple, consistent, but may not be efficient

Maximum Likelihood (MLE)

Likelihood: $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$

Log-likelihood: $\ell(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$

MLE: $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta)$

Finding MLE: Solve $\frac{d\ell}{d\theta} = 0$ (score equation)

MLE Properties:

- Invariance: If $\hat{\theta}$ is MLE of θ , then $g(\hat{\theta})$ is MLE of $g(\theta)$
- Consistent under regularity conditions
- Asymptotically efficient
- Asymptotic normality: $\hat{\theta}_n \approx N(\theta, \frac{1}{nI(\theta)})$

Key MLE Examples:

- Bernoulli(p): $\hat{p} = \bar{X}$
- Poisson(λ): $\hat{\lambda} = \bar{X}$
- $U(0, \theta)$: $\hat{\theta} = \max\{X_1, \dots, X_n\} = X_{(n)}$ (boundary case)
- $N(\mu, \sigma^2)$: $\hat{\mu} = \bar{X}, \hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ (biased)

4. Confidence Intervals

Definition

$(1 - \alpha)100\%$ CI for θ : Random interval $[L, U]$ with

$$P(L \leq \theta \leq U) = 1 - \alpha$$

Interpretation: $(1 - \alpha)100\%$ of intervals contain true θ .

Pivotal Quantity Method:

- Find pivotal $Q(X, \theta)$ with known distribution (indep of θ)
- Find constants: $P(a \leq Q \leq b) = 1 - \alpha$
- Invert to get: $P(L(X) \leq \theta \leq U(X)) = 1 - \alpha$

Common CIs

Mean (σ known, large n):

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Mean (σ unknown, Normal):

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

Proportion (large n):

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Variance (Normal):

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}}$$

Ratio of Variances:

$$\frac{s_1^2/s_2^2}{F_{\alpha/2, n_1-1, n_2-1}} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} \cdot F_{\alpha/2, n_1-1, n_2-1}$$

Two-Sample Mean (equal var):

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$

Sample Size Determination

For mean with margin E and confidence $(1 - \alpha)$:

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2$$

For proportion:

$$n = \hat{p}(1 - \hat{p}) \left(\frac{z_{\alpha/2}}{E} \right)^2$$

Conservative (no prior $\hat{p}$): Use $\hat{p} = 0.5$ (gives largest n)

5. Hypothesis Testing

Framework

Null hypothesis H_0 : Status quo (assume true)

Alternative H_1 : What we want evidence for

Types:

- Two-sided: $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$
- Upper: $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$

- Lower: $H_0 : \theta \geq \theta_0$ vs $H_1 : \theta < \theta_0$

Error Types

Type I Error (False Positive):

$\alpha = P(\text{Reject } H_0 | H_0 \text{ true})$

Type II Error (False Negative):

$\beta = P(\text{Fail reject } H_0 | H_1 \text{ true})$

Power: $1 - \beta = P(\text{Reject } H_0 | H_1 \text{ true})$

p-Value

Probability under H_0 of observing test statistic as extreme or more extreme.

Decision: Reject H_0 if $p \leq \alpha$

Interpretation:

- $p \leq 0.01$: Very strong evidence against H_0
- $0.01 < p \leq 0.05$: Moderate evidence
- $0.05 < p \leq 0.10$: Weak evidence
- $p > 0.10$: Little/no evidence

Common Test Statistics

Z-test (mean, σ known):

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1) \text{ under } H_0$$

t-test (mean, σ unknown):

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ under } H_0$$

Proportion test (large n):

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0, 1) \text{ under } H_0$$

Chi-square (variance):

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \sim \chi_{n-1}^2 \text{ under } H_0$$

Two-sample t-test (equal var):

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2} \text{ under } H_0$$

F-test (variance ratio):

$$F = \frac{s_1^2}{s_2^2} \sim F_{n_1-1, n_2-1} \text{ under } H_0 : \sigma_1^2 = \sigma_2^2$$

Rejection Regions

Two-sided:

- Z or t: $|T| > t_{\alpha/2}$

$$\chi^2: \chi^2 < \chi_{1-\alpha/2}^2 \text{ or } \chi^2 > \chi_{\alpha/2}^2$$

$$F: F < F_{1-\alpha/2} \text{ or } F > F_{\alpha/2}$$

One-sided (upper):

$$Z \text{ or } t: T > t_\alpha$$

$$\chi^2: \chi^2 > \chi_\alpha^2$$

$$F: F > F_\alpha$$

Power & Sample Size

$$\text{Power: } \pi(\mu_1) = \Phi\left(\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} - z_\alpha\right)$$

$$\text{Sample size: } n = \left(\frac{(z_\alpha + z_\beta)\sigma}{\mu_1 - \mu_0}\right)^2$$

Two-sided: Use $z_{\alpha/2}$ instead of z_α

CI-Testing Duality

If $(1 - \alpha)$ CI for θ is $[L, U]$:

- $\theta_0 \notin [L, U]$: Reject $H_0 : \theta = \theta_0$ at level α
- $\theta_0 \in [L, U]$: Fail to reject H_0

6. Likelihood Ratio Tests

Neyman-Pearson Lemma

For simple $H_0 : \theta = \theta_0$ vs simple $H_1 : \theta = \theta_1$:

Most powerful test of size α rejects when:

$$\Lambda(x) = \frac{L(\theta_0; x)}{L(\theta_1; x)} \leq k$$

where k satisfies $P(\Lambda \leq k | H_0) = \alpha$

General LRT

For composite hypotheses $H_0 : \theta \in \Theta_0$ vs

$H_1 : \theta \in \Theta_1$:

Parameter Spaces:

- Ω = full parameter space (unrestricted)
- Ω_0 = restricted parameter space under H_0

Likelihood ratio:

$$\lambda(x) = \frac{\sup_{\theta \in \Omega_0} L(\theta; x)}{\sup_{\theta \in \Omega} L(\theta; x)} = \frac{L(\hat{\theta}_0; x)}{L(\hat{\theta}; x)}$$

where $\hat{\theta}_0$ = MLE under H_0 (restricted), $\hat{\theta}$ = MLE (unrestricted)

Properties: $0 \leq \lambda(x) \leq 1$, smaller $\lambda \rightarrow$ stronger evidence against H_0

Wilks' Theorem

Under H_0 and regularity conditions:

$$-2 \log \lambda(X) \xrightarrow{d} \chi_\nu^2$$

where $\nu = \dim(\Theta) - \dim(\Theta_0)$ (df = difference in parameters)

Rejection: $-2 \log \lambda(x) > \chi_{\alpha, \nu}^2$

7. Key Distributions Summary

Dist	Mean	Var	Support
$\chi^2(k)$	k	$2k$	$(0, \infty)$
$t(n)$	0	$\frac{n}{n-2}$	$(-\infty, \infty)$
$F(n_1, n_2)$	$\frac{n_2}{n_2-2}$	-	$(0, \infty)$

Relationships:

- $\sum_{i=1}^k Z_i^2 \sim \chi^2(k)$, $Z_i \sim N(0, 1)$ iid
- $\frac{Z}{\sqrt{V/n}} \sim t(n)$, $Z \sim N(0, 1)$, $V \sim \chi^2(n)$
- $\frac{V_1/n_1}{V_2/n_2} \sim F(n_1, n_2)$, $V_i \sim \chi^2(n_i)$
- $T^2 \sim F(1, n)$ if $T \sim t(n)$
- $\chi^2(k_1) + \chi^2(k_2) = \chi^2(k_1 + k_2)$ (independent)

8. Important Formulas

Sample Statistics (Normal):

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

Two Samples (Normal, equal var):

$$\frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1)$$

MLE Asymptotic Normality:

$$\hat{\theta}_{\text{MLE}} \approx N\left(\theta, \sigma_{\text{MLE}}^2\right) \text{ for large } n$$

Delta Method:

$$g(\hat{\theta}_n) \approx N\left(g(\theta), [g'(\theta)]^2 \text{Var}(\hat{\theta}_n)\right)$$

LRT Asymptotic:

$$-2 \log \lambda \xrightarrow{d} \chi_\nu^2$$